PRECALCULUS
FINAL PRACTICE REVIEW

NAME:

Question 1. Which of the following functions satisfy the definition of a linear map? *You do not need to justify your answer, but space is provided.*

(a) $f(x) = 3x$

(b) $f(x) = x + 2022$

(c) $f(x) = \cos(x)$

(d) $f(x) = -\dfrac{x}{2}$

Recall if v, w are vectors and f is a linear map then

$$f(v + w) = f(v) + f(w), \quad f(cv) = c \cdot f(v)$$

where c is a number/constant.

(a) $f(x) = 3x$ yes

(b) $f(x) = x + 2022$ no if x and y are vectors then

$$f(x+y) = (x+y) + 2022$$

$$f(x) + f(y) = x + 2022 + y + 2022$$

(c) $f(x) = \cos(x)$ no

(d) $f(x) = -\dfrac{x}{2}$ yes

Question 2. Suppose $f : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear map that such that

$$f(1,2) = (-3,4), \qquad f(3,4) = (1,-1).$$

What is $f(20,22)$?

Recall the definition of the linear map:

if v, w are vectors and a, b are numbers/constants, then f is a linear map if

(1) $f(v+w) = f(v) + f(w)$
(2) $f(av) = af(v)$

and this must be true for every vector and constant.

Since we know the outputs of the vectors $v = (1,2)$ and $w = (3,4)$, i.e., we know what $f(v)$ and $f(w)$ are from the question.

So we need to find constants a, b such that $av + bw = a(1,2) + b(3,4) = (20,22)$. Using the vector operations (vector addition and scalar multiplication with vectors)

$$\begin{aligned} av + bw = a(1,2) + b(3,4) = (20,22) &\implies (a, 2a) + (3b, 4b) = (20,22) \\ &\implies (a+3b, 2a+4b) = (20,22) \\ &\implies \begin{cases} a + 3b = 20 \\ 2a + 4b = 22 \end{cases} \end{aligned}$$

and so we need to solve for a and b. You may use any method you want (substitution, elimination, matrix reduction, etc.) and you should conclude that $a = -7, b = 9$.

So now to solve the problem:

$$\begin{aligned} f(20,22) &= f(av+bw) = f(av) + f(bw) = af(v) + bf(w) \\ &= -7f(1,2) + 9f(3,4) \\ &= -7(-3,4) + 9(1,-1) && \text{using problem prompts} \\ &= (21,-28) + (9,-9) = (30,-38) \end{aligned}$$

so $f(20,22) = (30,-38)$.

Question 3. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ and $g : \mathbb{R}^2 \to \mathbb{R}^2$ be the function given by the equations

$$f(x,y) = (y,-x) \qquad \text{and} \qquad g(x,y) = (y,x).$$

(a) Let A_f be the matrix representation of f and let A_g be the matrix representation of g in standard bases. Write down the matrices A_f and A_g.

Standard basis for $\mathbb{R}^2$ are: $(1,0)$ and $(0,1)$

$$f(1,0) = (0,-1) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$f(0,1) = (1,0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$A_f = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$g(1,0) = (0,1) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$g(0,1) = (1,0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$A_g = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

(b) Compute $\det(A_f)$ and $\det(A_g)$ as well as the product $\det(A_f) \cdot \det(A_g)$.

$$\det(A_f) = \det\left(\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\right) = 0 - (-1) = 1$$

$$\det\big(A_g\big) = \det\left(\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\right) = 0 - (1) = -1$$

(c) Compute the matrix $A_g \cdot A_f$.

$$A_g \cdot A_f = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \qquad = \begin{bmatrix} 0 \cdot 0 + 1 \cdot (-1) & 0(1) + 1(0) \\ 1 \cdot 0 + 0 \cdot (-1) & 1(1) + 0(0) \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

(d) Find the equation for the composition $g \circ f : \mathbb{R}^2 \to \mathbb{R}^2$.

$$f(x, y) = (y, -x), \qquad g(x, y) = (y, x)$$

$$\begin{aligned} (g \circ f)\underbrace{(x, y)}_{v} &= g\big(f(x, y)\big) \\ &= g(y, -x) && \text{def of } f \\ &= (-x, y) && \text{def of } g \end{aligned}$$

(e) Find the matrix representation of the linear map $g \circ f$, we'll denote this as $A_{g \circ f}$.
Given that $(g \circ f)(x, y) = (-x, y)$ use the standard bases:

$$(g \circ f)(1, 0) = (-1, 0) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$(g \circ f)(0, 1) = (0, 1) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

so $A_{g \circ f} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

(f) Compute $\det\big(A_{g \circ f}\big)$.

$$\det\left(\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}\right) = -1$$

(g) Confirm that $A_{g\circ f} = A_g \cdot A_f$ and $\det\big(A_{g\circ f}\big) = \det\big(A_g\big) \cdot \det\big(A_f\big)$.
They match! Groovy.

(h) (Bonus) Compute the matrix $A_f \cdot A_g$ and compare this to your answer in (d). How do they compare? (i.e., are they equal?)

Question 4. Let $v = (1, 0, 0)$ and $w = (0, 1, 0)$.

(a) Compute the cross product $v \times w$.
For specifically $\mathbb{R}^3$, we have $i = (1, 0, 0)$ and $j = (0, 1, 0)$ and $k = (0, 0, 1)$ and to compute the cross product, we take the determinant of

$$v \times w = \det\left(\begin{bmatrix} i & j & k \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix}\right)$$

where $v = (v_1, v_2, v_3)$ and $w = (w_1, w_2, w_3)$.
So in our specific case, we have $v = (1, 0, 0)$ and $w = (0, 1, 0)$ so

$$v \times w = \det\left(\begin{bmatrix} i & j & k \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}\right) = k$$

so $v \times w = (0, 0, 1)$.

(b) Are v and w orthogonal? Recall that the v and w have their inner product equalling 0, where inner product is given by

$$\langle v, w\rangle = v_1 \cdot w_1 + v_2 w_2 + v_3 w_3 + \cdots + v_n w_n$$

where $v = (v_1, v_2, \ldots, v_n), w = (w_1, w_2, \ldots, w_n) \in \mathbb{R}^n$.
Applying this to our problem we have that $v = (1, 0, 0)$ and $w = (0, 1, 0)$ so we compute the inner product:

$$\langle v, w\rangle = 1(0) + 0(1) + 0(0) = 0$$

so v and w are orthogonal to each other (they're perpendicular to each other).

(c) Compute the cross product $w \times v$.
So in our specific case, we have $v = (1, 0, 0)$ and $w = (0, 1, 0)$ so

$$w \times v = \det\left(\begin{bmatrix} i & j & k \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}\right) = -k$$

so $v \times w = (0, 0, -1)$.

(d) Are the vectors $v \times w$ and $w \times v$ orthogonal to each other?
Since $v \times w = (0, 0, 1)$ and $w \times v = (0, 0, -1)$, we can compute the inner products:

$$\langle (0, 0, 1), (0, 0, -1)\rangle = 0(0) + 0(0) + 1(-1) = -1$$

and since their inner product is **not** zero, they are **not** orthogonal.

Question 5. Solve for x in the equation $9^x - 10 \cdot 3^x + 9 = 0$.

Let $u = 3^x$. Then it follows that $u^2 = (3^x)^2 = 3^{2x}$.

Noting that $9^x = (9)^x = (3^2)^x = 3^{2x}$, we can rewrite the given equation as

$$u^2 - 10u + 9 = 0$$

which is a quadratic! We simply need to solve for u and then retranslate it into x at the very end.

Solving for the roots of u:

$$u^2 - 10u + 9 = 0 \Longrightarrow (u-9)(u-1) = 0 \Longrightarrow u = 9, 1$$

and so we conclude that $3^x = 9$ and $3^x = 1$.

Solving for x, we get $x = 2, 0$.

Question 6. Let $x > 0$ and let a_n be a geometric sequence with

$$a_1 = \ln\left(x^9\right), \quad a_2 = \ln\left(x^3\right), \quad a_3 = \ln(x).$$

(a) Find the common ratio.
(b) Recall that the sum of an infinite geometric sequence is given by the formula

$$\frac{firstterm}{1 - ratio}, \qquad |ratio| < 1$$

Given that $\sum_{n=1}^{\infty} a_n = 54$, where a_n is our given sequence, compute the value of x.

(a) Recall that to find the common ratio of a geometric sequence that we have to divide a given term by the previous term, i.e., $R = a_2/a_1$. So we get that

$$R = \frac{a_2}{a_1} = \frac{\ln\left(x^3\right)}{\ln\left(x^9\right)}$$

We are not able to proceed any further without recalling rules of logarithms:
From the power rule of logarithms, which says

$$\log_b(x^n) = n \log_b(x)$$

we can simplify R to

$$R = \frac{\ln\left(x^3\right)}{\ln\left(x^9\right)} = \frac{3\ln(x)}{9\ln(x)} = \frac{3}{9} = \frac{1}{3}.$$

(b) Since we have that $|R| < 1$, we have that the infinite sum has a limit which can be computed by

$$\sum_{n=1}^{\infty} a_n = \frac{a_1}{1 - R}$$

and so plugging in what we have, we conclude that the infite sum must be

$$\sum_{n=1}^{\infty} a_n = \frac{9\ln(x)}{1-(1/3)} = \frac{9\ln(x)}{2/3} = \frac{9\ln(x)}{1}\cdot\frac{3}{2} = \frac{27\ln(x)}{2}.$$

Using the problem prompt, which says that the sum is equivalent to 54, we get that

$$\frac{27\ln(x)}{2} = 54 \Longrightarrow \ln(x) = \frac{2\cdot 54}{27} = 4$$

and so we get that $\ln(x) = 4 \Longrightarrow x = e^4$ (since ln has base e).

Question 7. Let $\sinh(x) = \dfrac{e^x - e^{-x}}{2}$ and $\cosh(x) = \dfrac{e^x + e^{-x}}{2}$.

(a) Prove that $\frac{\mathrm{d}}{\mathrm{d}x}\sinh(x) = \cosh(x)$
(b) Prove that $\frac{\mathrm{d}}{\mathrm{d}x}\cosh(x) = \sinh(x)$.

Recall that the chain rule says

$$(f\circ u)'(x) = f'(u(x))\cdot u'(x)$$

and so in summary this says

$$\frac{\mathrm{d}}{\mathrm{d}x}e^{f(x)} = e^{f(x)}\cdot f'(x), \qquad \frac{\mathrm{d}}{\mathrm{d}x}\sin\big(f(x)\big)$$
$$\frac{\mathrm{d}}{\mathrm{d}x}f(x)^n = n(f(x))^{n-1}\cdot f'(x), \qquad \frac{\mathrm{d}}{\mathrm{d}x}\sqrt{f(x)} = \frac{1}{2\sqrt{f(x)}}\cdot f'(x), etc.$$

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\sinh(x) &= \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{e^x - e^{-x}}{2}\right] \\
&= \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{1}{2}\left(e^x - e^{-x}\right)\right] \\
&= \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}x}\left[e^x - e^{-x}\right] \\
&= \frac{1}{2}\left(\frac{\mathrm{d}}{\mathrm{d}x}e^x - \frac{\mathrm{d}}{\mathrm{d}x}e^{-x}\right) && \text{sum/difference rule} \\
&= \frac{1}{2}\left(e^x - e^u\cdot u'(x)\right) && \text{where } u = -x, \text{using chain rule} \\
&= \frac{1}{2}\left(e^x - e^{-x}\cdot(-1)\right) \\
&= \frac{1}{2}\left(e^x + e^{-x}\right) = \cosh(x)
\end{aligned}$$

Similarly,

$$\frac{\mathrm{d}}{\mathrm{d}x}\cosh(x) = \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{e^x + e^{-x}}{2}\right]$$

$$
\begin{aligned}
&= \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{1}{2}\left(e^x + e^{-x}\right)\right] \\
&= \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}x}\left[e^x + e^{-x}\right] \\
&= \frac{1}{2}\left(\frac{\mathrm{d}}{\mathrm{d}x}e^x + \frac{\mathrm{d}}{\mathrm{d}x}e^{-x}\right) && \text{sum/difference rule} \\
&= \frac{1}{2}\left(e^x + e^u \cdot u'(x)\right) && \text{where } u = -x, \text{using chain rule} \\
&= \frac{1}{2}\left(e^x + e^{-x} \cdot (-1)\right) \\
&= \frac{1}{2}\left(e^x - e^{-x}\right) = \sinh(x)
\end{aligned}
$$

Question 8. How many 11-character strings can we form from the string `yippeekayay`

The total would be 11! if all of the letters were unique, but there are repeats!

There are:

(1) y appears 3 times
(2) p appears 2 times
(3) e appears 2 times
(4) a appears 2 times
(5) k, i appears once,

So the total number characters is

$$\frac{11!}{3!2!2!2!}$$

Question 9. How many different groups of 8 can we form from a group of 12?

This is computed by $\binom{12}{8}$ (12 choose 8) and is computed by the fraction

$$\binom{12}{8} = \frac{12!}{8!(12-8)!} = \frac{12!}{8!4!} = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8!}{8! \cdot 4!} = \frac{12 \cdot 11 \cdot 10 \cdot 9}{4 \cdot 3 \cdot 2 \cdot 1}$$

Question 10. Let $\mathbb{R}[x]_{\leq n}$ denote the set of all (real) polynomials with degree n or less. For example, $\mathbb{R}[x]_{\leq 3}$ consists of all (real) polynomials with degree 3 or less.

Let $d : \mathbb{R}[x]_{\leq 3} \to \mathbb{R}[x]_{\leq 2}$ denote the derivative and let $f(x) = 1 + x + x^2 + x^3$ and $g(x) = 1 - 2x + 3x^2 - 4x^3$.

(a) Show that $d(2f(x)) = 2d(f(x))$.
(b) Show that $d(f(x) + g(x)) = d(f(x)) + d(g(x))$
(c) Is d a linear map between the two spaces?
(d) Give the matrix representation of the derivative with respect to the standard bases.

I will only answer part (c) and (d). You can use the derivative rules to prove (a) and (b).

Since (a) and (b) are true, the function d is a linear map! This means we can give a matrix representation:

Let $1 = (1, 0, 0, 0)$ in the domain space and $1 = (1, 0, 0)$ in the output space and similarly, we have $x = (0, 1, 0, 0), x^2 = (0, 0, 1, 0), x^3 = (0, 0, 0, 1)$ in the domain and $x = (0, 1, 0)$ and $x^2 = (0, 0, 1)$ in the output spaces (noting that the domain is 4 dimensional while outputs are 3 dimensions).

Since $\{1, x, x^2, x^3\}$ is the standard basis for the domain and $\{1, x, x^2\}$ is the standard basis for the output space, we compute the output of each standard basis vector:

$$\mathrm{d}(1,0,0,0) = d(1) = 0 = (0,0,0) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathrm{d}(0,1,0,0) = d(x) = 1 = (1,0,0) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathrm{d}(0,0,1,0) = d(x^2) = 2x = (0,2,0) = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$$

$$\mathrm{d}(0,0,0,1) = d(x^3) = 3x^2 = (0,0,3) = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$$

and so the matrix representation for $\mathrm{d} : \mathbb{R}[x]_{\geq 3} \to \mathbb{R}[x]_{\geq 2}$ is given by

$$\mathrm{d} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

Question 11. Let $z = 3 + 3\sqrt{3}i$.

(a) Compute $|z|$. Compute the magnitude by

$$|z| = \sqrt{v_1^2 + v_2^2} = \sqrt{3^2 + (3\sqrt{3})^2} = \sqrt{9 + 27} = 6$$

(b) Compute $\arg(z)$.

To compute $\arg(z)$, by writing $z = x + yi$ and get the argument by computing

$$\theta = \arg(z) = \arctan\big(y/x\big) = \arctan\left(\frac{3\sqrt{3}}{3}\right) = \arctan\Big(\sqrt{3}\Big) =$$

Converting z into polar coordinates, we get that $z = 6\left(\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right)$. Using De Moivre's theorem, which says

$$z^n = r^n\left(\cos(n\theta) + i\sin(n\theta)\right)$$

we get that

$$z^{2022} = 6^{2022}\left(\cos\left(\frac{2022\pi}{3}\right) + i\sin\left(\frac{2022\pi}{3}\right)\right).$$

Question 12. Suppose $v = (1, 2, 3, 4)$ and $w = (1, -3, 5, -7)$ is a vector.

(a) What dimension does v live in?
(b) Compute $|v|$.
(c) Are v and w orthogonal?

(1) v is in dimension 4
(2) $|v|$ is computed by

$$|v| = \sqrt{v_1^2 + v_2^2 + v_3^2 + v_4^2} = \sqrt{1^2 + 2^2 + 3^2 + 4^2} = \sqrt{1 + 4 + 9 + 16} = \sqrt{30}$$

(3) To see if v and w are orthgonal, compute the inner product and see if it's zero:

$$\begin{aligned}\langle v, w\rangle &= \langle (1, 2, 3, 4), (1, -3, 5, -7)\rangle \\ &= 1(1) + 2(-3) + 3(5) + 4(-7) \\ &= 1 - 6 + 15 - 28 = -18 \neq 0\end{aligned}$$

and so they are not orthogonal.

Question 13. Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

(a) Compute $\det(A)$.
(b) Does A^{-1} exist? If so, find it. If not, state why.
(c) Compute A^2 and A^3.
(d) What is A^n? (assuming $n \in \mathbb{Z}$). Prove your result by induction for all $n \geq 1$.

(a) For a 2×2 matrix, we compute determinant by

$$\det\left(\begin{bmatrix} a & c \\ b & d \end{bmatrix}\right) = ad - bc$$

and so $\det(A) = 1 - 0 = 1$.

(b) Yes, A^{-1} exists since $\det(A) \neq 0$. To compute the inverse, we want to solve for the identity matrix:

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \to R_1 - R_2} \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & 0 & 1 \end{array}\right]$$

and since the left half is the identity matrix, we have solved for our inverse of A, given by the right half:

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}.$$

(c) Recall matrix multiplication

`https://youtu.be/kT4Mp9EdVqs`

So to compute A^2, we simply multiply A with A and since the width of the left matches the height of the right matrix, we can multiply them:

$$A^2 = AA = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

Similarly, we compute A^3 :

$$A^3 = A^2 A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

(d) We might suspect that A^n where $n \in \mathbb{Z}$ is

$$A^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}.$$

Let's prove it by induction (for $n \geq 1$).

Proof. We first do a base case and then we do the inductive case.

(a) Base case: Let $n = 1$. Then we get that

$$A^1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

and for the right half of the equality, if we set $n = 1$, we get the exact same matrix and so our base case is true.

(b) Inductive case: Suppose our claim is true for some $n \geq 1$, i.e., suppose that the following equality is true:

$$A^n = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}.$$

We will show that A^{n+1} will be exactly $\begin{bmatrix} 1 & n+1 \\ 0 & 1 \end{bmatrix}$:

$$A^{n+1} = A^n A = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & n+1 \\ 0 & 1 \end{bmatrix}$$

and so we have finished our inductive case.

By the power of POMI, our claim is true for all $n \geq 1$.[1] $\square$

Question 14. Find the limit of $\lim_{x \to 0} \dfrac{\sin(2x)}{x}$.

You may either use L'Hopital's rule or you can use limit rules and bending the problem to rewrite the probelm into

$$\lim_{x \to 0} \frac{2}{1} \frac{\sin(2x)}{2x} = 2 \cdot \lim_{x \to 0} \frac{\sin(2x)}{2x} = 2 \cdot 1 = 2$$

Question 15. Compute the limit of $\lim_{x \to 0} \dfrac{1 - \cos(2x)}{4x}$

You may either use L'Hopital's rule or you can use limit rules and bending the problem to rewrite the probelm into

$$\lim_{x \to 0} \frac{1}{2} \frac{1 - \cos(2x)}{2x} = \frac{1}{2} \cdot \lim_{x \to 0} \frac{1 - \cos(2x)}{2x} = \frac{1}{2} \cdot 0 = 0$$

[1] Remark: it is easy to show that it is true for $n = 0$. Moreover, it is also easy to show it is true for all $n < 0$ since we have already proven the base case for $n = -1$; we just need to do an induction argument assuming it is true for some n and proving is also true for $n - 1$. This proves it for all $n \in \mathbb{Z}$.